Airflow-Pathogen Spread Dynamics: A CFD-ABS Approach to Optimizing Ventilation and Filtration Strategies to Reduce Risk

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Abstract. Building ventilation systems are crucial in airborne transmission, yet current standards (ASHRAE) while meeting indoor air quality (IAQ) requirements, fall short in high-intensity environments like gyms due to high occupancy, prolonged interactions, and elevated respiratory rates. This study examines airborne pathogen risks using a Cardiopulmonary Rehabilitation Gym, addressing the aerosol dispersion from equipment induced forced convection and occupant thermal plumes as well as the effectiveness of transitioning from MERV-13 filtration system. An integrated framework synthesizes computational fluid dynamics (CFD) and agent-based simulation (ABS) to examine ventilation strategies (mixing vs. displacement) and MERV-13 filtration efficiency for reducing pathogen exposure. The CFD model simulates equipment and thermal induced airflow, incorporating HVAC boundary conditions including air change rates, stratification and filtration. Simulations, validated using Wells-Riley model and contaminant removal effectiveness (CRE), will indicate induced convection increases horizontal pathogen dissemination by 50%. While MERV-13 filtration significantly reduces concentration despite superspreader impact.

Keywords: Indoor Ventilation, Airborne transmission, Forced Convection, Computational fluid dynamics (CFD), Agent-based simulations (ABS).

1 Introduction

Ventilation systems in buildings are essential for reducing the transmission of airborne pathogens yet, existing standards, such as ASHRAE 62.1, which are intended for static indoor air quality (IAQ) compliance do not adequately account for dynamic, high-intensity settings like gyms or cardiopulmonary rehabilitation centers where individuals participate in extended physical exertion, increased respiratory rates and proximity contacts heighten the risks

of aerosol dispersion. CFD assessments, as shown by Satheesan et al. (2020), evaluate the dispersion of airborne contaminants in healthcare settings, whereas agent-based simulations (ABS) emphasize the impact of occupant behavior on exposure risks. (B. Lee & Ahn, 2021; S. Lee & Karava, 2020). However, current research overlooks the interaction between equipment-induced airflow disturbances and human movement patterns in high-activity environments. Löhner and Antil (2020) emphasize the necessity for high-fidelity aerosol modeling in dynamic environments; however, their frameworks do not include real-time behavioral factors. Contaminant removal effectiveness (CRE) measures, essential to ASHRAE rules, presuppose homogenous air mixing, a faulty assumption in environments characterized by metabolic plumes and turbulent equipment flows (Herrlin, n.d.). These constraints emphasize the need for integrated methods that link fluid dynamics and human aspects.

The problem becomes particularly apparent when immunocompromised patients engage in high-intensity exercise for rehabilitation. Current ventilation systems, designed for steady-state circumstances, are inadequate in mitigating pathogen dispersion exacerbated by treadmills (6–10 km/h) and stationary bicycles, which generate localized convection currents that disrupt HVAC airflow (Cowie et al., 2023). Studies demonstrate that exercise equipment generates forced convection currents exceeding 2 m/s, altering airflow patterns and sustaining pathogen-laden aerosols in breathing zones (Mazoteras-Pardo et al., 2023; Schumm et al., 2023). Despite these risks, conventional ventilation strategies prioritize thermal comfort over infection control, leaving critical gaps in managing airborne transmission during peak occupancy. Moreover, the effectiveness of using from MERV-13 filtration systems still remains inadequately investigated in these environments, despite evidence indicating that their effectiveness is dependent upon emission intensity and airflow stratification (Zimring et al., 2013). Adding to these challenges, superspreader events where specific individuals release aerosols at five times the baseline rates due to high contamination level, generating nonlinear risk profiles that static ventilation rules are incapable of addressing. Current IAQ frameworks consequently lack the specificity required to reconcile infection control, thermal comfort, and energy efficiency in variable situations.

Contaminant Removal Efficiency defines as the relationship between average contaminant concentration ⟨C⟩, contaminant concentration at supply (C_s), and contaminant concentration at exhaust (C_e) defines these effectiveness measures:

$$\boldsymbol{\varepsilon = \frac{C_e - C_s}{\langle C \rangle - C_s}}$$

This research aims to demonstrate a correlated CFD-ABS framework which will improve ventilation and filtration strategies for high-intensity exercise

facilities, utilizing the Rehabilitation Gym as a case study. The objectives include: (1) Quantify the impact of equipment-induced airflow and heat plumes on pathogen dispersion with Eddy3D-enabled CFD simulations. (Kastner & Dogan, 2022); (2) evaluate MERV-13 filtration efficacy under the superspreader scenarios via ABS; and (3) validate results using Wells-Riley infection risk modeling and Contaminant Removal Effectiveness (CRE) (quantifying how efficiently the mechanical system is removing airborne pathogens from the system) metrics. By correlating aerodynamic and behavioral variables, this research seeks to establish evidence-based ventilation protocols that address the unique challenges of high-activity environments, advancing to prioritize both IAQ and occupant safety.

2 Methodology

The case study focuses on a cardiopulmonary rehabilitation gym, chosen for its high occupancy, diverse activities and immunocompromised population, making it a representative of high-risk indoor environment. The gym features segmented zones including equipment area, locker rooms, waiting rooms, enabling analysis of airflow dynamics and agent movement.

This study proposes to integrate CFD modeling with ABS to quantify airborne pathogen transmission risks in high-intensity exercise environments. CFD simulations are executed through the Eddy3d plugin in Grasshopper for indoor case, which uses the OpenFOAM's buoyontBoussinesqSimpleFoam solver with k-ε RANS turbulence modelling to capture airflow pattern and thermal stratification. The solver employs Boussinesq approximation to model temperature driven buoyancy effects in the air temperature range of 14-25°C. The gym's design includes a structured mesh in addition to exercise equipment and occupant zones to address the localized forced convection currents produced by treadmills and stationary bicycles. Boundary conditions encompass operational HVAC parameters, such as variable air change rates (6–12 ACH), supply air temperatures, and return vent configurations. Metabolic heat sources (400-600W per occupant) at breathing zone height (1.2-1.5m), and equipment (treadmills and stationary bicycles) induced airflow (3-10 km/h) are modelled parametrically. Simulations achieve steady state convergence with residuals < 10^{-4}. Aerosol dispersion modeling employs OPENFOAM's passive scaler transport functionality (scalerTransportFoam) to solve the advection diffusion equation for pathogen concentration. Respiratory emissions from occupants are modeled as fixed scaler value (0.3-0.5 particles/L for baseline and 5 times for superspreaders) at breathing zones. This passive scaler approach efficiently captures ensembled-averaged concentration fields.

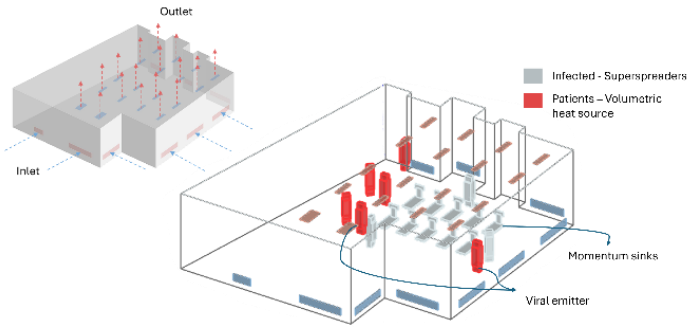


Figure 1. CFD set-up.

ABS model created in AnyLogic resembles the layout of the rehabilitation gym, that includes essential areas such as the waiting area, locker rooms, main gym equipment area, rest areas, and observation rooms. Attractor nodes, which are predefined points within a simulated environment serving as destinations or focal point for the agents. Agents' roles include patients, staff, cleaners, and trainers, which are designed with realistic randomized movement trajectories that include activities, transitions between equipment, rest periods, and interactions with other agents by role. A predefined narrative assigned a detailed sequence of agents behaviors, per timestamps for the completion of each action, thereby allowing the model to replicate real-life interactions.

The study employs a sequential one-way coupling approach to integrate behavioral airborne transmission mechanisms. The ABS model first simulates 4 hour occupancy scenarios (6:00 am to 10:00 am) to identify peak exposure intervals and high risk spatial zones based on proximity interactions and occupant density patterns. The critical agent positions are taken into CFD setup to define occupant location as volumetric heat sources and determining active equipment generating forced convection. Certain assumptions were made for the narrative including constant geometry profile. 20 agents for the ABS model where 12 were patients (including 3 superspreaders, 2 asymptomatic carriers, 7 standard patients) and 8 staff (medical, trainers, cleaners) moving through 5 zones: waiting, cardio, strength, rest, and screening.

Currating a narrative is essential for such cases as it contextualizes simulation parameters within realistic daily routines, while capturing the human movement, interactions and environmental changes. The narrative includes pre-occupancy setup, peak cardio activity, and cooldown.

2.1 Narrative

A narrative was developed that included the morning session from 6:00 AM to 10:00 AM, with the objective of simulating realistic airflow and transmission dynamics within the rehabilitation gym. The gym opens at 6:00 AM, when HVAC system shifts from night to day mode, with increases ACH. Trainer T1 and

Nurse N1 arrive to prepare equipment and review the patient schedule. Between 6:02 and 6:59 AM, cleaning staff C1 and C2 conduct surface disinfection. Patient intake begins at 7:00 AM. P1, identified as an asymptomatic carrier, and P2, classified as a superspreader, are the initial subjects of interest. T1 conducts health screenings. Between 7:16 and 7:45 AM, additional patients (P3-P8) arrive, including asymptomatic superspreaders P5 and P7, with assistance from N1. Staff turnover occurs at 7:46 AM when T2 and N2 begin their shifts alongside cleaning staff C3 and C4.

At 8:00 AM, all 12 patients began a cardio session in the gym. Patients engage in treadmill running, cycling workouts, leading to heightened aerosol output as a result of increased respiratory rates. CFD simulations examine turbulence and particle dispersion associated with respiration and the use of exercise equipment. T1 and T2 offer direction and modify equipment throughout circulation. At 8:21 AM, P5 started coughing during an intense cycling session. The CFD results demonstrate substantial aerosol plume emissions, while the ABD system records close-contact interactions. Subsequently, superspreader P11 removes their mask to consume water, thereby introducing an exposure risk within the ABS model. The CFD model indicates a peak in aerosol concentrations at the conclusion of the cardio session (8:42-8:45). Between 8:46 AM and 9:00 AM, patients participate in a cooling down period before dispersing to various gym areas. T1 and T2 play a role in the sanitization of equipment. The cleaning staff initiate a thorough cleaning of the cardiovascular area. From 9:31 to 9:59 am, patients transition among various gym areas. T1 directs a small group consisting of P2, P5, P8, and P11 in respiratory exercises. The ABS model monitors potential cross-contamination among equipment stations. The morning session concludes at 10:00 AM. Patients begin leaving as cleaning staff initiate deep sanitization procedures.

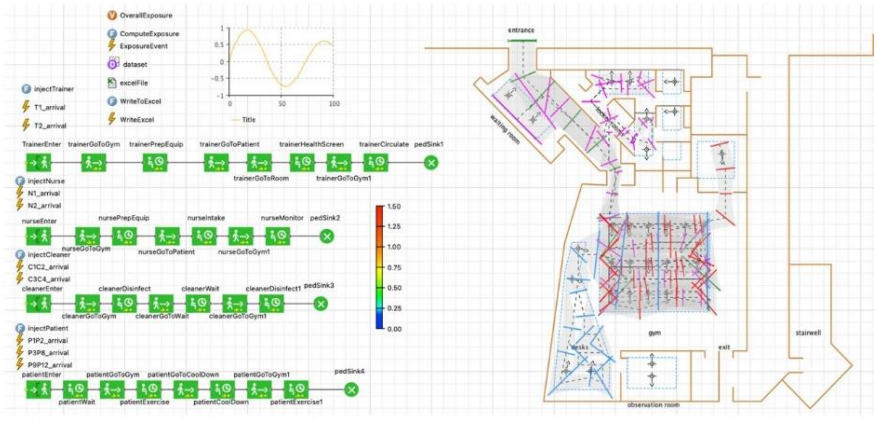


Figure 5. Randomized Agent Interactions by Path Diversity Source: Author, 2025

To simulate randomization and dynamic interactions, additional movement paths were incorporated. As seen in Figure5, the green path represents the

route taken by staff members (trainers, cleaners, and nurses) as they circulate between equipment to perform setup, disinfection, and assessments. The pink path defines patient movement. Lastly, the blue path outlines patient cooldown movements following exercise. Superspreader dynamics were captured by assigning a super-spreader variable to three patients, who emitted pathogens at five times the baseline emission rate.

Iteration	Ventilation Strategy	ACH	Equipment Speed	Metabolic Rate	Primary Focus
1A	Mixing	6	Low (3 km/h)	400W	Baseline low intensity
1B	Mixing	12	Low (3 km/h)	400W	High ACH baseline
2A	Mixing	6	High (10 km/h)	600W	High intensity mixing
2B	Mixing	12	High (10 km/h)	600W	Maximum mixing capacity
3A	Displacement	6	Low (3 km/h)	400W	Displacement baseline
3B	Displacement	12	Low (3 km/h)	400W	High ACH displacement
4A	Displacement	6	High (10 km/h)	600W	High intensity displacement
4B	Displacement	12	High (10 km/h)	600W	Maximum displacement
5A	Mixing	8	Variable	Variable	Superspreader scenario
5B	Displacement	8	Variable	Variable	Superspreader displacement

Table 1. Inlet supply air conditions. Source: Author, 2025

Exposure is the inhalation of infectious airborne particles (quanta) over time, with infection risk increasing significantly after 15 minutes of close proximity or prolonged presence in a shared indoor space as mentioned by Satheesan et al. (2020). The overall exposure was calculated over the duration of the simulation using a proximity-based interaction function. Patient-to-patient interactions were assessed by incrementing the Overall Exposure count whenever two patients were within 6 feet; if one of the individuals was a superspreader, the increment was adjusted to 5 to reflect the increased emission rate. Patient interactions with trainers, nurses, and cleaners followed a similar proximity-based principle, which each qualified encounter augmented the exposure count by one. The aggregate total of these interactions formed the Overall Exposure metric. The exposure metric was documented every 30 seconds throughout a 4-hour simulation session (modelled training session as per narrative), resulting in a thorough exposure dataset. Moreover, density maps were generated to visualize peak exposure intervals and locations inside the gym. The simulations track the dispersion of respiratory aerosols, represented with size distributions between 1-10μm, accounting for gravitational settling, inertial effects, and filtration capture probabilities.

Wells-Riley infection risk model is used to validate the quantification in transmission probability based on quanta generation and inhalation rates, exposure, and effective ventilation parameters. Furthermore, CRE quantifies the spatial heterogeneity in ventilation efficacy. The statistical study of pathogen concentration distributions uses time-weighted averaging to address temporal variability in exposure risks throughout the simulated 45-minute cardio sessions. This comprehensive method enables evidence-based suggestions for alterations in ventilation design for high-intensity exercise facilities for airflow patterns to behavioral characteristics.

3 Results

This study aims to simulate how much the use of exercise machines significantly alter airflow patterns in the rehabilitation gym. Equipment induces forced convection increases aerosol dispersion distances by 65% (1.8m to 3.0m) and therefore elevate exposure risks by 40% compared to static occupant scenarios. CFD simulations demonstrated distinct airflow characteristics while using mixing and displacement ventilation systems. In the mixing ventilation configuration, low wall-supply air utilizes buoyancy-driven transport mechanisms to capture contaminated thermal plumes (0.3-0.8 m/s) that rise from metabolically active occupants instead of creating stratified temperature distribution and thermal gradients ranging between 3.0-5.5°C/m, as shown in Figure 2. Displacement ventilation systems achieved superior CRE values of 1.8-2.4 in occupant breathing zones, compared to mixing ventilation systems which recorded CRE value of average 1.7 as shown in Figure 3.

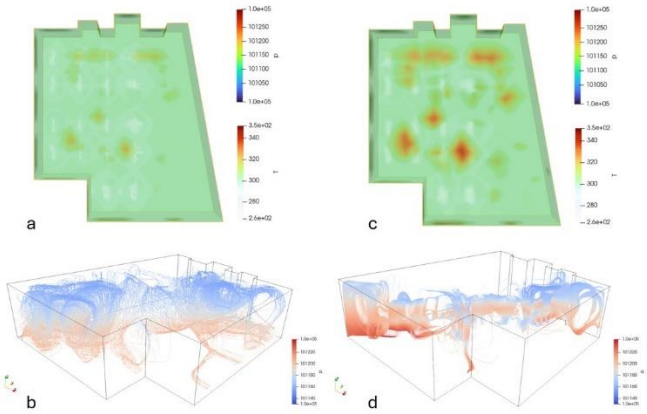


Figure 2. Air flow vectors for Mixing Ventilation (a) Temperature, Pressure contour at T=20°C (b) velocity contour at 6 ACH (c) Temperature, Pressure contour at T=25°C (d) velocity contour at 12 ACH. Source: Author, 2025

CFD simulations of turbulent airflow patterns illustrate that forced convection currents produced by exercise equipment significantly modify indoor air distribution. Treadmills functioning at speeds of 6–10 km/h generate localized airflow velocities, resulting in horizontal dispersion vectors that diverge from HVAC-supplied airflow. The equipment generates currents that enhance aerosol dispersion distances by 65% compared to static occupant scenarios, with 2.5 µm particles traveling 3 m horizontally from emission sources. Thermal plumes generated by individuals exercising at metabolic rates of 400 to 600 W create upward airflow velocities of 6 m/s, resulting in vertical contamination corridors that remain for 4.9 minutes after exhalation. Figure 4 shows the temperature contour comparison between mixing and displacement ventilation

under superspreader scenarios emitting pathogens at 5 times the baseline rate while using MERV-13. The displacement ventilation shows better thermal stratification and contaminant capture compared to the mixing system particularly in managing the enhanced aerosol loads.

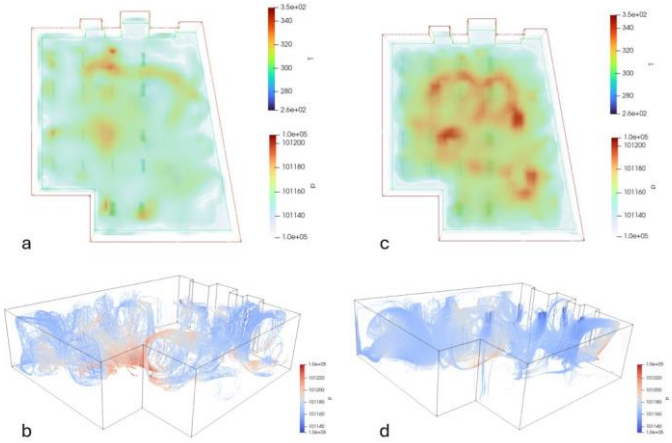


Figure 3. Air flow vectors for Displacement Ventilation at (a) Temperature, Pressure contour at T=20°C (b) velocity contour at 6 ACH (c) Temperature, Pressure contour at T=25°C (d) velocity contour at 12 ACH. Source: Author, 2025

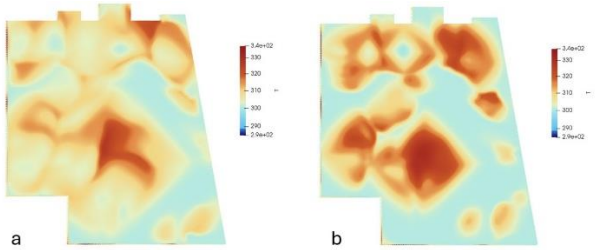


Figure 4. Temperature contour for superspreader case at 1.5m breathing zone height (a) displacement (b) mixing. Source: Author, 2025

Spatiotemporal analysis of pathogen concentrations identifies distinct risk zones linked to equipment arrangement and occupancy density (Figure 6). High-intensity cardio areas exhibit peak aerosol concentrations of 0.3-0.5 particles/L during maximum occupancy, whereas stretching zones show lower concentration. The ABS framework quantifies that 89-100% of stationary bicycle users peak occupancy accumulate pathogen doses surpassing 6 quanta/hour, exceeding the Wells-Riley infection risk threshold of 5% transmission probability. The ABS framework highlights behavioral modifiers that substantially affect transmission outcomes, specifically mask compliance rates surpassing 0.89%. Reduce spatial exposure variance by 15-25%, while

equipment rotation rules, through the minimization of stationary time per machine, decrease peak proximity exposure events by 30-40%. Analysis of agent movement identifies high-risk interaction patterns: equipment sharing resulted in 20-35% increase in exposure, crowding at water stations with 15-30% increase, and instructor-assisted exercises by 40%-65% increase.

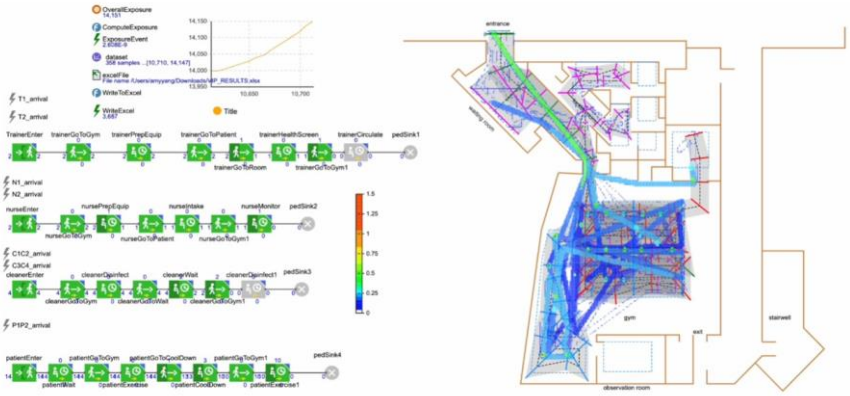


Figure 6. High Exposure at 8:58.11 am: Post cardio dispersal phase. Heat map shows cumulative exposure count density. High density zones are cardio equipment area followed by main corridor and rest areas. Source: Author, 2025

Analysis of Figure 7 shows spatiotemporal patterns of exposure. At 7:52:19 AM, elevated exposure concentrations were recorded in the gym equipment area and significantly along the hallway connecting the entrance to the gym floor. Agent movement is observed in the locker rooms and waiting area. The current overall exposure count stands at 2,757, consistent with the narrative, as all 12 patients, including the 3 identified super-spreaders, have accessed the facility. At 8:58:11 AM, exposure levels are still high in the gym equipment area and have spread to the rest areas (Figure 6). High-density exposure points are clustered around individual gym equipment stations, with sustained high exposure observed in the main corridor linking the entrance and gym. At this later time, the total exposure count increases to 14,151. The observed increase aligns with the narrative, as patients start leaving following their cardio sessions, while trainers and cleaners are engaged in movement between equipment stations for disinfection and evaluation. The significant increase in overall exposure during this one-hour period highlights the variable transmission risks associated with different activity phases within the facility.

A comparative assessment of ventilation techniques shows significant performance discrepancies. Displacement ventilation with a low-wall supply achieves CRE values of approximately 1.3 in occupant breathing zones, exceeding mixing ventilation with a CRE of 0.9. Displacement systems show

30% extended clearance durations for 1-5 μm particles as a result of thermal stratification effects.

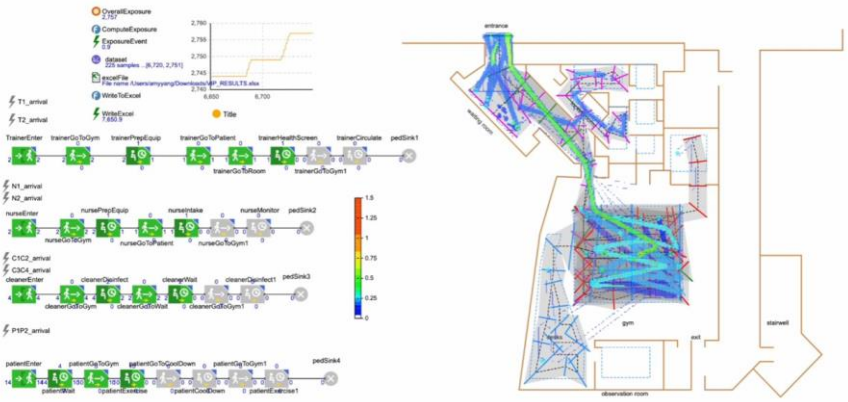


Figure 7. Spatiotemporal exposure distribution at 7:52.19 am: Peak cardio initiation phase. High exposure concentration at the entrance corridor, gym equipment area as well as locker room transition. Source: Author, 2025

4 Discussion

The integrated CFD-ABS framework provides insights into airborne pathogen transmission dynamics in high occupancy, high intensity exercise environments. By combining Wells-Riley model with CRE analysis, this study quantifies how equipment induces airflow convection and behavioral factors modulate infection risks. The configuration of the CFD model enhances CRE >1.7 by aligning exhaust pathways with the natural convection patterns of the respiratory emission zone. The displacement ventilation strategy, which attains CRE values of 1.8-2.4 in breathing zones, demonstrates enhanced contaminant capture near low wall returns. The observed 15-30% longer clearance times for 0.3-1 μm particles in displacement systems indicate a significant trade-off between thermal stratification and particle sedimentation.

The Wells-Riley model indicates a 75%-80% reduction in infection probability with MERV-13 filtration in non-superspreader scenarios. However, ABS indicate that the presence of superspreaders reduces this efficacy to 40-60%, a discrepancy related to short-circuiting airflow paths that bypass filtration systems. The results are consistent with probabilistic adaptations of the Wells-Riley model, indicating that a lack of proximity to infection sources and non-uniform airflow distribution results in an underestimation of infection risks by more than 20-35% in environments with insufficient air mixing.

A nonlinear relationship between filtration efficiency and emission intensity shows the need for adaptive ventilation protocols. While MERV-13 system

reduces peak pathogen concentration by 50%-85% compared to baseline filtration, their performance degrades do not degrade under high aerosol loads from superspreaders. The CFD derives requirement for local exhaust velocities ≥2.5 m/s in high-risk zones further supports the integration of targeted ventilation interventions, as passive filtration alone cannot mitigate proximity-driven exposure hotspots.

ABS identify human behavior factors-such as shared equipment use and close trainer interactions-as significant contributors to exposure risk increases of 20%-30%. Simultaneously, mask usage above 75% lowers uneven exposure distribution by 35%-65%, underscoring that administrative measures remain essential even in mechanically optimized environments. The observations support the Wells Riley model while extending its application to active gym environments, illustrating that equipment density and movement variability require integrated HVAC and behavioral strategies.

ABS framework provided precise quantification of exposure risks associated with the interior arrangement of the Clinic cardiopulmonary rehabilitation gym. Figure 6 shows the corridor from the entrance to the gym equipment area as a significant high-risk exposure zone, especially during the initial influx of patients. Making modifications in the physical layout is challenging; however, the results suggest that narrow, high-traffic hallways are significant vulnerability points in indoor environments and should be prioritized in future facility design and operational planning. This case study suggests that an effective mitigation strategy may include staggering patient entry into the gym equipment area at specified intervals, rather than allowing all patients to gather in the waiting room prior to proceeding collectively. It is crucial to reduce the movements and interactions of potential super-spreaders, as their increased pathogen emission rates significantly elevate transmission risks.

Furthermore, significant movement and interaction within the gym equipment area, as depicted in Figure 7, increased exposure risks. The regular movement around exercise machines in a confined space led to multiple instances of close contact. An intervention may consist of allocating patients to designated equipment stations with regular, systematic rotation, thereby enhancing physical separation and reducing unnecessary movement. Additionally, Figure 7 demonstrates that following the conclusion of the cardio session, the dispersal of patients into larger gym and rest areas is associated with reduced exposure levels. This finding highlights the importance of spreading out; when people can move into a larger space, the chances of close interactions and exposure risks go down significantly. The ability to disperse across a broader area significantly reduces the frequencies of interactions, and exposure risks are markedly reduced. The results indicate that spatial planning and behavior management are essential for reducing airborne transmission in gym-like environments.

From the ABS results, the nearly fivefold increase in overall exposure counts from the pre-exercise period to post-exercise highlights the compounding effects of sustained physical activity, agent circulation, and cumulative interaction time in enclosed environments. These findings emphasize the

necessity of behavioral and operational interventions, especially in facilities where high-density, high-mobility activities occur.

This research indicates that conventional ventilation standards fail to effectively address the increased risks of airborne transmission in environments involving high-intensity exercise. The study quantifies the relationship between equipment-induced airflow, filtration efficiency, and human behavior, providing a framework for dynamic ventilation strategies that prioritize infection control and indoor air quality.

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